

Paper A-10: Earth Sustainability: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

Founder, Vura Labs

Kampala, Uganda

ORCID: 0009-0001-0556-5516

May 2026

Abstract

This paper develops a falsifiable CDFD/AFL model of Earth Sustainability. It maps human demand for energy, materials, food, land, and welfare to driving flux Φ , regenerative capacity, pollution sinks, institutions, and distribution to active constraint C , innovation, substitution, repair, governance, and demand adjustment to responsiveness S , and infrastructure lock-in, ecological degradation, wealth, and institutional history to structural memory M_s . The target outcome is durable welfare within environmental and social limits, tested against material-flow accounts, life-cycle inventories, ecological indicators, and welfare series. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow-constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine

[5]. Part IV [6], DOI <https://doi.org/10.5281/zenodo.20395901>, is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [3], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \rightarrow \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

The Earth is a closed thermodynamic system with respect to matter, and an open system with respect to solar energy. Human civilization currently operates on a linear extraction model: pulling matter from the lithosphere, processing it with fossil energy, and dumping it into the atmosphere and oceans.

This linear model inherently requires exponential growth. Under CDFD, this equates to demanding infinite flux (Φ) against finite constraints (C). Physics strictly forbids this. The system will enter an overload regime.

3 The Mathematics of Survival

We evaluate the long-term viability of the biosphere using the Universal Adaptive Ratio:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

For a sustainable global civilization:

- **Flux (Φ):** The total mass-energy throughput of civilization. To be sustainable, this flux must shift entirely to renewable, closed-loop sources (solar, wind, circular recycling).
- **Constraint (C):** The absolute physical limits of the Earth: available land, raw elemental abundance, and the thermal dissipation limit of the atmosphere.
- **Responsiveness (S):** The efficiency of technology and the agility of societal governance. A sustainable society possesses extreme technological responsiveness, allowing it to generate high economic utility from minimal physical flux.

- **Memory (M_s):** Ecological restoration and cultural wisdom. A sustainable society locks in beneficial memory (e.g., deep-rooted regenerative agriculture) to buffer the system against random external shocks.

3.1 The Steady-State Attractor ($\Psi_s = 1$)

Current macroeconomic models demand perpetual GDP growth, forcing the system into a dangerous over-flux regime ($\Psi_s \gg 1$). To achieve sustainability, humanity must intentionally enter the Steady-State Attractor.

In this regime, the raw physical flux (Φ) is capped. It cannot exceed the planetary constraint (C). One candidate route to improve human well-being without raising physical throughput is to increase S through technological, social, and institutional efficiency. We must decouple economic value from thermodynamic throughput.

By locking the system at exactly $\Psi_s = 1$, civilization aligns its metabolism perfectly with the Earth’s natural regenerative cycles. The global topological surface ceases to degrade, and the threat of a cascading planetary phase shift (collapse) is reduced under the declared conditions.

4 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system’s ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

4.1 Current Runtime Diagnostic: Universal Network Cascade

The release-local discovery script keeps this paper tied to the current CDFD Runtime [7], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the 2026-06-04 runtime pass, the localized hub-drive stress test ended with hub $\Psi_s = 5.21$, peak network $\Psi_s = 5.21$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

5 Major Universal Upgrade: Mechanism, Scale, and Tests

5.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Earth Sustainability, the proposed drive is human demand for energy, materials, food, land, and welfare. The active limitation is regenerative capacity, pollution sinks, institutions, and distribution. Responsiveness is carried by innovation, substitution, repair, governance, and demand adjustment, while retained state is represented by infrastructure lock-in, ecological

degradation, wealth, and institutional history. The outcome to explain is durable welfare within environmental and social limits.

Table 1: Operational CDFD/AFL map for Paper A-10.

Term	Paper-specific observable meaning	Measurement question
Φ	human demand for energy, materials, food, land, and welfare	What rate, load, gradient, or demand enters the focal system?
C	regenerative capacity, pollution sinks, institutions, and distribution	Which measured bottleneck limits transfer, function, or recovery?
S	innovation, substitution, repair, governance, and demand adjustment	Which process changes effective capacity or reroutes the load?
M_s	infrastructure lock-in, ecological degradation, wealth, and institutional history	Which retained state makes equal present loads produce different futures?
Y	durable welfare within environmental and social limits	Which outcome is predicted out of sample, with uncertainty?

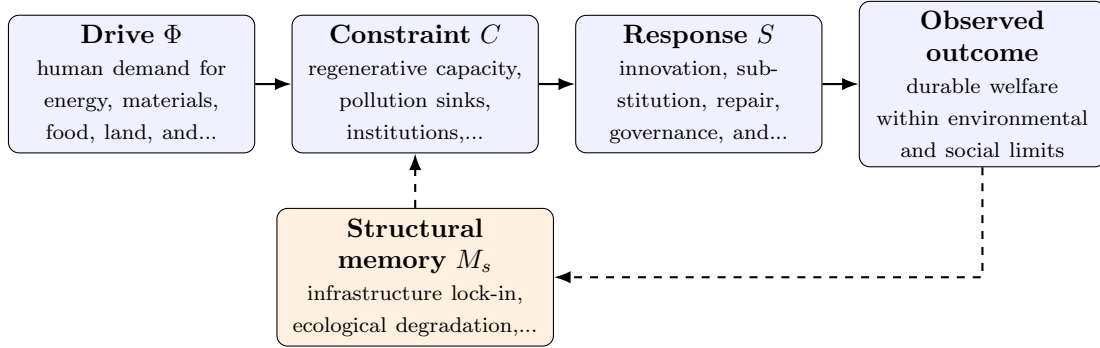


Figure 1: Paper A-10 observable CDFD/AFL architecture for Earth Sustainability. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

5.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \quad (2)$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \quad (3)$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \quad (4)$$

$$\Psi_s(t) = \frac{\hat{\Phi}(t)}{\epsilon + \hat{C}(t)} \hat{S}(t) [1 + \omega \hat{M}_s(t)]. \quad (5)$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in human demand for energy, materials, food, land, and welfare arrives faster than innovation, substitution, repair,

governance, and demand adjustment can compensate. This raises or redistributes regenerative capacity, pollution sinks, institutions, and distribution. If the load persists, the retained state encoded by infrastructure lock-in, ecological degradation, wealth, and institutional history changes the next response, so a later event of equal magnitude can produce a different trajectory in durable welfare within environmental and social limits. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

5.3 Cross-Scale Universality Without Mechanistic Erasure

For the Earth systems family, the universal comparison is between coupled transport, finite environmental capacity, buffering, and retained landscape or ecosystem state. Universality here cannot erase conservation laws, geometry, or scale; it must expose where those domain equations create comparable overload and recovery structures. [2, 8, 1] The CDFD programme is cumulative: Part I supplies the flow-constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is years to centuries; communities to the planet. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

5.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper A-10. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure material-flow accounts, life-cycle inventories, ecological indicators, and welfare series and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a demand increase combined with a capacity loss while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: the framework cannot outperform established sustainability accounting or identify trade-offs.

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recov-

ery phase estimates β and μ , separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

5.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from material-flow accounts, life-cycle inventories, ecological indicators, and welfare series and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of durable welfare within environmental and social limits. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

6 Conclusion

Sustainability is not an ideological preference; it is a mathematical boundary condition. By applying CDFD, we establish the absolute thermodynamic limits of human expansion. Achieving a sustainable future requires abandoning the pursuit of infinite flux (Φ) and instead mastering the topological elasticity (S) required to thrive within the unyielding constraints of our planetary surface.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

7 Reference Layer

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